

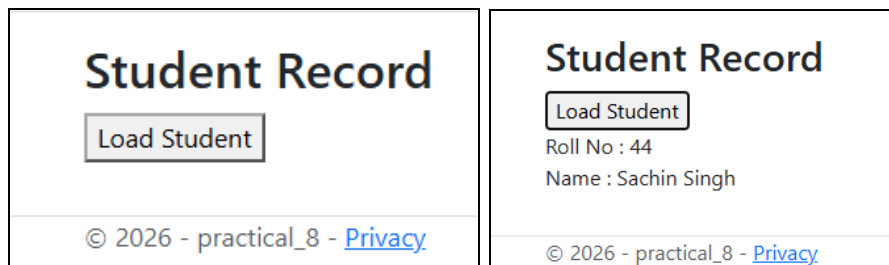
Practical – VIII

Roll No.: T044	Name: Sachin Singh
Class: TY.IT	Batch: 1
Date of Assignment: 06/08/2026	Date/Time of Submission: 06/08/2026

a. Create an HTML page that would be the starting point of the application. This HTML will contain a link to the file named manifest.json. This is an important file that would be created in the next step.

Index.cshtml

```
<h2>Student Record</h2>
<button onclick="loadStudent()">Load Student</button>
<p id="data"></p>
<script>
  function loadStudent() {
    fetch("/api/student")
      .then(response => response.json())
      .then(student => {
        document.getElementById("data").innerHTML =
          "Roll No : " + student.rollNo +
          "<br>Name : " + student.name;
      });
  }
</script>
```

**_Layout.cshtml**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - practical_8</title>
  <script type="importmap"></script>
  <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
  <link rel="stylesheet" href="~/practical_8.styles.css" asp-append-version="true" />
  <link rel="manifest" href="/manifest.json">_
</head>
<body>
```

Sachin Singh T044

```
<header>
  <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom
box-shadow mb-3">
    <div class="container-fluid">
      <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">practical_8</a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
aria-expanded="false" aria-label="Toggle navigation">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
        <ul class="navbar-nav flex-grow-1">
          <li class="nav-item">
            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
          </li>
          <li class="nav-item">
            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
          </li>
        </ul>
      </div>
    </div>
  </nav>
</header>
<div class="container">
  <main role="main" class="pb-3">
    @RenderBody()
  </main>
</div>
<footer class="border-top footer text-muted">
  <div class="container">
    &copy; 2026 - practical_8 - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
  </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
</body>
</html>
```

Manifest.json

```
{
  "name": "Student PWA",
  "short_name": "Student",
  "start_url": "/",
  "display": "standalone"
```

Student Record

Load Student

Roll No : 44

Name : Sachin Singh

© 2026 - practical_8 - [Privacy](#)

b. Create a simple Web API that returns Student data in JSON format.

Student.cs

```
namespace StudentPWA.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

StudentController.cs

```
using Microsoft.AspNetCore.Mvc;
namespace StudentPWA.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class StudentController : ControllerBase
    {
        [HttpGet]
        public IActionResult Get()
        {
            return Ok(new
            {
                RollNo = 44,
                Name = "Sachin Singh"
            });
        }
    }
}
```

